Machine Learning Assignment :- 6

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Use the diabetes data set from UCI and Pima Indians Diabetes data set for performing the following:

a. Univariate analysis: Frequency, Mean, Median, Mode, Variance, Standard Deviation, Skewness and Kurtosis

b. Bivariate analysis: Linear and logistic regression modeling

c. Multiple Regression analysis

d. Also compare the results of the above analysis for the two data sets.

DATASET LINK :- https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from scipy.stats import skew, kurtosis
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.metrics import r2_score, mean_squared_error, accuracy_score, confusion_matrix
import statsmodels.api as sm
```

```
data = pd.read_csv("diabetes.csv")
print("Original Shape:", data.shape)
data.head()
```

```
Original Shape: (768, 9)
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1

Next steps: Generate code with data New interactive sheet

```
zero_columns = ["Glucose", "BloodPressure", "SkinThickness", "Insulin", "BMI"]

for col in zero_columns:
    data[col] = data[col].replace(0, np.nan)

clean_data = data.dropna()

print("Cleaned Shape:", clean_data.shape)
```

```
Cleaned Shape: (392, 9)
```

A. Univariate Analysis: Frequency, Mean, Median, Mode, Variance, Standard Deviation, Skewness and Kurtosis

```
print("Outcome Distribution:")
print(clean_data["Outcome"].value_counts())
```

```
Outcome Distribution:
Outcome
0    262
1    130
Name: count, dtype: int64
```

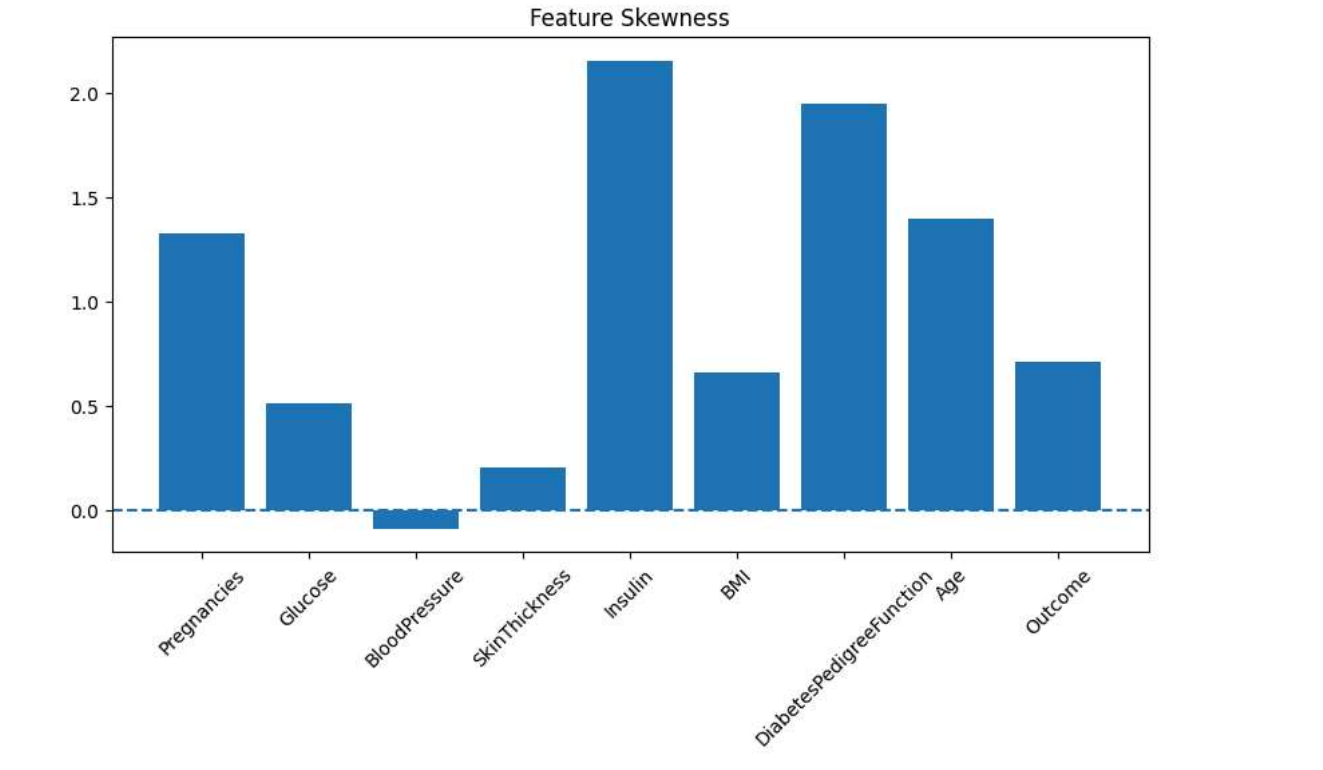
```
uni_stats = pd.DataFrame()

uni_stats["Mean"] = clean_data.mean()
uni_stats["Median"] = clean_data.median()
uni_stats["Mode"] = clean_data.mode().iloc[0]
uni_stats["Variance"] = clean_data.var()
uni_stats["Std_Deviation"] = clean_data.std()
uni_stats["Skewness"] = clean_data.apply(skew)
uni_stats["Kurtosis"] = clean_data.apply(kurtosis)

print("Univariate Analysis Summary")
display(uni_stats)
```

Univariate Analysis Summary

	Mean	Median	Mode	Variance	Std_Deviation	Skewness	Kurtosis
Pregnancies	3.301020	2.0000	1.000	10.313247	3.211424	1.330480	1.452184
Glucose	122.627551	119.0000	100.000	952.387781	30.860781	0.515866	-0.492353
BloodPressure	70.663265	70.0000	70.000	156.152304	12.496092	-0.087181	0.770028
SkinThickness	29.145408	29.0000	32.000	110.595171	10.516424	0.208509	-0.467146
Insulin	156.056122	125.5000	105.000	14123.347226	118.841690	2.156822	6.260449
BMI	33.086224	33.2000	32.000	49.387994	7.027659	0.660944	1.521463
DiabetesPedigreeFunction	0.523046	0.4495	0.692	0.119362	0.345488	1.951597	6.270504
Age	30.864796	27.0000	22.000	104.055842	10.200777	1.398230	1.700180
Outcome	0.331633	0.0000	0.000	0.222219	0.471401	0.715240	-1.488432

Next steps: Generate code with uni_stats New interactive sheet

```
plt.figure(figsize=(10,5))
plt.bar(uni_stats.index, uni_stats["Skewness"])
plt.xticks(rotation=45)
plt.axhline(0, linestyle="--")
plt.title("Feature Skewness")
plt.show()
```



B. Bivariate Analysis: Linear and logistic regression modeling

```
correlation_matrix = clean_data.corr()

print("Correlation Matrix:")
print(correlation_matrix)
```

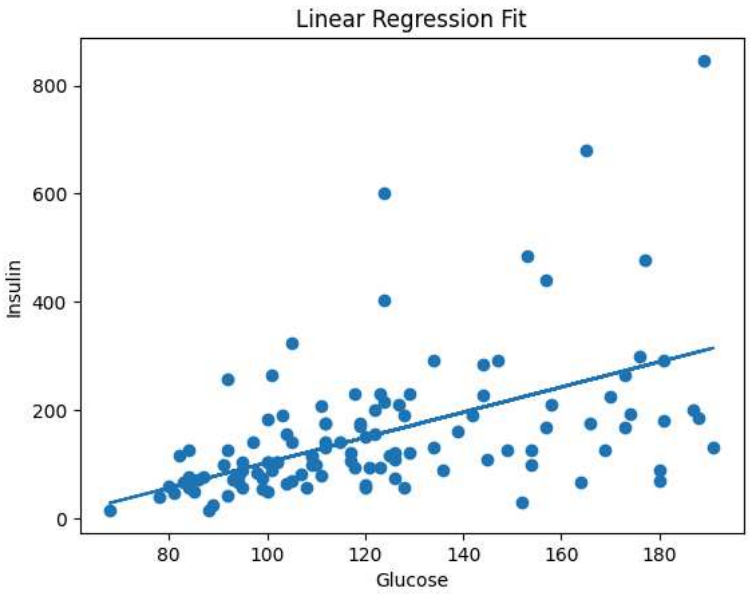
```
Correlation Matrix:
                         Pregnancies   Glucose  BloodPressure  SkinThickness  \
Pregnancies                 1.000000  0.198291       0.213355       0.093209
Glucose                     0.198291  1.000000       0.210027       0.198856
BloodPressure               0.213355  0.210027       1.000000       0.232571
SkinThickness               0.093209  0.198856       0.232571       1.000000
Insulin                     0.078984  0.581223       0.098512       0.182199
BMI                        -0.025347  0.209516       0.304403       0.664355
DiabetesPedigreeFunction    0.007562  0.140180      -0.015971       0.160499
Age                         0.679608  0.343641       0.300039       0.167761
Outcome                     0.256566  0.515703       0.192673       0.255936

                           Insulin       BMI  DiabetesPedigreeFunction  \
Pregnancies               0.078984 -0.025347                  0.007562
Glucose                   0.581223  0.209516                  0.140180
BloodPressure             0.098512  0.304403                 -0.015971
SkinThickness             0.182199  0.664355                  0.160499
Insulin                   1.000000  0.226397                  0.135906
BMI                       0.226397  1.000000                  0.158771
DiabetesPedigreeFunction  0.135906  0.158771                  1.000000
Age                       0.217082  0.069814                  0.085029
Outcome                   0.301429  0.270118                  0.209330

                               Age   Outcome
Pregnancies               0.679608  0.256566
Glucose                   0.343641  0.515703
BloodPressure             0.300039  0.192673
SkinThickness             0.167761  0.255936
Insulin                   0.217082  0.301429
BMI                       0.069814  0.270118
DiabetesPedigreeFunction  0.085029  0.209330
Age                       1.000000  0.350804
Outcome                   0.350804  1.000000
```

```
X_lr = clean_data[["Glucose"]]
y_lr = clean_data["Insulin"]

X_train, X_test, y_train, y_test = train_test_split(
    X_lr, y_lr, test_size=0.3, random_state=1
)
```

```
linear_model = LinearRegression()
linear_model.fit(X_train, y_train)

predictions = linear_model.predict(X_test)

print("Linear Regression Output")
print("Slope:", linear_model.coef_[0])
print("Intercept:", linear_model.intercept_)
print("R2 Score:", r2_score(y_test, predictions))
print("MSE:", mean_squared_error(y_test, predictions))
```

```
Linear Regression Output
Slope: 2.327936397806107
Intercept: -129.91364653340716
R2 Score: 0.2272405309300114
MSE: 12603.30002639569
```

```
plt.scatter(X_test, y_test)
plt.plot(X_test, predictions)
plt.xlabel("Glucose")
plt.ylabel("Insulin")
plt.title("Linear Regression Fit")
plt.show()
```




```
X_log = clean_data.drop("Outcome", axis=1)
y_log = clean_data["Outcome"]

X_train, X_test, y_train, y_test = train_test_split(
    X_log, y_log, test_size=0.3, random_state=1
)

log_model = LogisticRegression(max_iter=1000)
log_model.fit(X_train, y_train)

log_predictions = log_model.predict(X_test)

print("Logistic Regression Results")
print("Accuracy:", accuracy_score(y_test, log_predictions))
print("Confusion Matrix:\n", confusion_matrix(y_test, log_predictions))
```

```
Logistic Regression Results
Accuracy: 0.7542372881355932
Confusion Matrix:
 [[68 14]
 [15 21]]
```

C. Multiple Regression Analysis

```
X_multi = clean_data.drop("Outcome", axis=1)
y_multi = clean_data["Outcome"]

X_multi = sm.add_constant(X_multi)

multi_reg_model = sm.OLS(y_multi, X_multi).fit()

print(multi_reg_model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                Outcome   R-squared:                       0.346
Model:                            OLS   Adj. R-squared:                  0.332
Method:                 Least Squares   F-statistic:                     25.30
Date:                Fri, 27 Feb 2026   Prob (F-statistic):           2.65e-31
Time:                        16:29:58   Log-Likelihood:                -177.76
No. Observations:                 392   AIC:                             373.5
Df Residuals:                     383   BIC:                             409.3
Df Model:                           8
Covariance Type:            nonrobust
==============================================================================================
                                 coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------------------
const                         -1.1027      0.144     -7.681      0.000      -1.385      -0.820
Pregnancies                    0.0130      0.008      1.549      0.122      -0.003       0.029
Glucose                        0.0064      0.001      7.855      0.000       0.005       0.008
BloodPressure               5.465e-05      0.002      0.032      0.975      -0.003       0.003
SkinThickness                  0.0017      0.003      0.665      0.506      -0.003       0.007
Insulin                       -0.0001      0.000     -0.603      0.547      -0.001       0.000
BMI                            0.0093      0.004      2.391      0.017       0.002       0.017
DiabetesPedigreeFunction       0.1572      0.058      2.708      0.007       0.043       0.271
Age                            0.0059      0.003      2.109      0.036       0.000       0.011
==============================================================================
Omnibus:                        9.511   Durbin-Watson:                   1.920
Prob(Omnibus):                  0.009   Jarque-Bera (JB):                9.387
Skew:                           0.344   Prob(JB):                      0.00916
Kurtosis:                       2.682   Cond. No.                     1.77e+03
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.77e+03. This might indicate that there are
strong multicollinearity or other numerical problems.
```

D. Also compare the results of the above analysis for the two data sets.

```
uci_data = pd.read_csv("diabetes.csv")

print("UCI Dataset Shape:", uci_data.shape)
print("Pima Dataset Shape (Cleaned):", clean_data.shape)

if len(uci_data.columns) == len(clean_data.columns):
    print("both datasets have same features.")
else:
    print("Feature mismatch found.")
```

```
UCI Dataset Shape: (768, 9)
Pima Dataset Shape (Cleaned): (392, 9)
both datasets have same features.
```

```
print("UCI Outcome Distribution:")
print(uci_clean_data["Outcome"].value_counts())

uci_uni_stats = pd.DataFrame()
uci_uni_stats["Mean"] = uci_clean_data.mean()
uci_uni_stats["Median"] = uci_clean_data.median()
uci_uni_stats["Mode"] = uci_clean_data.mode().iloc[0]
uci_uni_stats["Variance"] = uci_clean_data.var()
uci_uni_stats["Std_Deviation"] = uci_clean_data.std()
uci_uni_stats["Skewness"] = uci_clean_data.apply(skew)
uci_uni_stats["Kurtosis"] = uci_clean_data.apply(kurtosis)
print("\nUCI Univariate Analysis Summary")
display(uci_uni_stats)
```

```
UCI Outcome Distribution:
Outcome
0    262
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UCI Univariate Analysis Summary
```

	Mean	Median	Mode	Variance	Std_Deviation	Skewness	Kurtosis
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Outcome	0.331633	0.0000	0.000	0.222219	0.471401	0.715240	-1.488432

Next steps: Generate code with uci_uni_stats New interactive sheet

```
uci_correlation_matrix = uci_clean_data.corr()

print("UCI Correlation Matrix:")
print(uci_correlation_matrix)
```

```
UCI Correlation Matrix:
                          Pregnancies   Glucose  BloodPressure  SkinThickness  \
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                           Insulin       BMI  DiabetesPedigreeFunction  \
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```